

SKILL-4

1. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

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rishitha@Rishitha: ~/skill4
rishitha@Rishitha:~$ mkdir skill4
rishitha@Rishitha:~$ cd skill4
rishitha@Rishitha:~/skill4$ pwd
/home/rishitha/skill4
rishitha@Rishitha:~/skill4$ nano lexer.c
rishitha@Rishitha:~/skill4$ gcc lexer.c -o lexer
rishitha@Rishitha:~/skill4$ ls
lexer  lexer.c
rishitha@Rishitha:~/skill4$ ./lexer
Enter command: ls -l /home

Tokens:
WORD: ls
WORD: -l
WORD: /home
rishitha@Rishitha:~/skill4$ ls -l |grep txt
rishitha@Rishitha:~/skill4$ touch text.txt
rishitha@Rishitha:~/skill4$ ls -l | grep txt
-rw-r--r-- 1 rishitha rishitha    0 Sep 18 08:30 text.txt
rishitha@Rishitha:~/skill4$ nano parser.c
rishitha@Rishitha:~/skill4$ gcc parser.c -o parser
rishitha@Rishitha:~/skill4$ ls
lexer  lexer.c  parser  parser.c  text.txt
rishitha@Rishitha:~/skill4$ ./parser
Enter command: ls -l
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
rishitha@Rishitha:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
rishitha@Rishitha:~/skill4$ ./parser
Enter command: | ls
Syntax Error: command expected before |
rishitha@Rishitha:~/skill4$ ls -l
total 40
-rwxr-xr-x 1 rishitha rishitha 16176 Sep 18 08:29 lexer
-rw-r--r-- 1 rishitha rishitha   813 Sep 18 08:29 lexer.c
-rwxr-xr-x 1 rishitha rishitha 16176 Sep 18 08:31 parser
-rw-r--r-- 1 rishitha rishitha  1057 Sep 18 08:30 parser.c
-rw-r--r-- 1 rishitha rishitha    0 Sep 18 08:30 text.txt
```

```
rishitha@Rishitha: ~/skill4
rishitha@Rishitha:~/skill4$ ls -l | grep txt
-rw-r--r-- 1 rishitha rishitha 0 Sep 18 08:30 text.txt
rishitha@Rishitha:~/skill4$ nano parser.c
rishitha@Rishitha:~/skill4$ gcc parser.c -o parser
rishitha@Rishitha:~/skill4$ ls
lexer lexer.c parser parser.c text.txt
rishitha@Rishitha:~/skill4$ ./parser
Enter command: ls -l
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
rishitha@Rishitha:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
rishitha@Rishitha:~/skill4$ ./parser
Enter command: | ls
Syntax Error: command expected before |
rishitha@Rishitha:~/skill4$ ls -l
total 40
-rwxr-xr-x 1 rishitha rishitha 16176 Sep 18 08:29 lexer
-rw-r--r-- 1 rishitha rishitha 813 Sep 18 08:29 lexer.c
-rwxr-xr-x 1 rishitha rishitha 16176 Sep 18 08:31 parser
-rw-r--r-- 1 rishitha rishitha 1057 Sep 18 08:30 parser.c
-rw-r--r-- 1 rishitha rishitha 0 Sep 18 08:30 text.txt
rishitha@Rishitha:~/skill4$ ./lexer
Enter command: ls -l | grep txt

Tokens:
WORD: ls
WORD: -l
DELIMITER: |
WORD: grep
WORD: txt
rishitha@Rishitha:~/skill4$ ./parser
Enter command: ls -l | grep txt
Syntax Valid
Parse Structure: COMMAND -> ARGUMENTS
rishitha@Rishitha:~/skill4$ ./parser
Enter command: | ls
Syntax Error: command expected before |
rishitha@Rishitha:~/skill4$ |
```